

Adding Two-Digit Numbers with Place-Value Strategies

Answer Key

Grade 2 Math

Quick Answers

- | | |
|--|---|
| 1. tens = 50, ones = 9, sum = 59 | 84 |
| 2. tens = 70, ones = 8, sum = 78 | 6. Mia's first sum = 64, the real sum = 63, — |
| 3. tens = 60, ones = 14, sum = 74 | 7. tens = 80, ones = 13, sum = 93 |
| 4. (a) ones in all = 12, (b) the sum = 62 | 8. the total = 85, — |
| 5. after the first jump = 76, where you land = | |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 2 Math

2.NBT.B.5 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 21 tagged checks.

Common wrong answers

3. *If they got 614: wrote the 14 ones next to the 6 tens without regrouping*

1. Add $34 + 25$ by breaking both numbers apart.

Solution: Break apart: $34 = 30 + 4$ and $25 = 20 + 5$. Tens: $30 + 20 =$

Ones: $4 + 5 =$

Together: $50 + 9 =$

50

9

59

2. Add $41 + 37$ by breaking both numbers apart.

Solution: Break apart: $41 = 40 + 1$ and $37 = 30 + 7$. Tens: $40 + 30 =$

Ones: $1 + 7 =$

Together: $70 + 8 =$

70

8

78

3. Add $26 + 48$ (the ones make more than a ten).

Solution: Tens: $20 + 40 =$

60

Ones: $6 + 8 =$

14

— that is one ten and 4 ones, so regroup it into the tens when combining: $60 + 14 = 60 + 10 + 4.$

74

4. Base-ten blocks for $37 + 25$: (a) cubes in all, (b) the sum.

Solution: (a) The cubes are the ones: $7 + 5 =$

12

(b) Twelve cubes trade into 1 rod and 2 cubes. Rods: $3 + 2 + 1 = 6$, so the sum is 6 rods and 2 cubes: $30 + 20 + 12 =$

62

5. Number-line jumps for $46 + 38$: $+30$, then $+8$.

Solution: Start at 46. The first jump adds the tens: $46 + 30 =$

76

The second jump adds the ones: $76 + 8 = 84$, so $46 + 38 =$

84

6. Mia's friendly-number strategy for $39 + 24$.

Solution: The friendly number 40 is easy to add: $40 + 24 =$

64

But 40 is one *more* than 39, so the sum is one too big — take the extra one away: $64 - 1 = 63$, so $39 + 24 =$

63

What a good explanation contains: Mia added 40 instead of 39, which is one too many, so she must take one away at the end to make it fair.

7. Place-value chart for $57 + 36$.

Solution: Tens row: $50 + 30 =$

80

Ones row: $7 + 6 =$

13

— one ten and 3 ones. Combine and regroup: $80 + 13 = 80 + 10 + 3.$

93

8. Challenge: $45 + 27 + 13$, and which pair to group first.

Solution: Look at the ones digits: 7 and 3 make a friendly ten, so group $27 + 13 = 40$ first. Then $45 + 40 = 85$, so $45 + 27 + 13 =$

85

What a good explanation contains: the pair 27 and 13 was grouped first because their ones digits (7 and 3) make ten, giving the round number 40, which is easy to add to 45. (Any grouping that names the friendly ten earns full credit.)